

Generalised Fibonacci Sequence & Linear Algebra

Asked 3 years, 8 months ago Modified 3 years, 7 months ago Viewed 204 times



1



Consider a generalised fibonacci G sequence $1, 1, 1, 3, 5, 9, 17 \dots$ that's created by summing the last 3 entries in the sequence together:

$$G_0 = 1, G_1 = 1, G_2 = 1 \text{ and } G_{n+1} = G_n + G_{n-1} + G_{n-2} \text{ for } n \geq 2.$$

1) Find a 3×3 matrix M such that, for any $k \geq 2$,

$$\begin{pmatrix} G_{k+1} \\ G_k \\ G_{k-1} \end{pmatrix} = M \begin{pmatrix} G_k \\ G_{k-1} \\ G_{k-2} \end{pmatrix}$$

I figured out that $M = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$ from the given equation.

2) Find a numerical value for G_{25}

With a bit of thinking, I came up with the equation

$$\begin{pmatrix} G_{k+1} \\ G_k \\ G_{k-1} \end{pmatrix} = M^k \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

Hence G_{25} would be obtained by the first element from M^{24} times $[1, 1, 1]$. Using a calculator, I got 3311233, which apparently is wrong. Was my equation above wrong? How could I approach this?

Find limit of $\frac{\ln G_n}{n}$ to 10 decimal places as n goes to ∞ .

I have no idea how to approach this. I thought of using eigenvalues and eigenvectors, but since the M I found only gives me one real value of 1.83929... I don't quite see how that's going to be useful. Any help would be really appreciated.

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asked May 14, 2020 at 15:36



singularity
309 2 8

▲ Hint: the characteristic polynomial $x^3 = x^2 + x + 1$ has one root of norm > 1 and two roots of norm < 1 . Also: the value you computed is G_{26} . – [lulu](#) May 14, 2020 at 15:40 ✎

▲ Well I guess I can calculate each root's normal, but I'm not sure how that helps me exactly. Do the normals equal the eigenvalues? – [singularity](#) May 14, 2020 at 16:08

▲ Since only one root, call it α , has a significant norm then we know that $G_n \sim \lambda \times \alpha^n$ for some constant λ (at least when n is large). If more than one root had norm greater than 1, no such simplification would be possible (at least for generic choices of

initial conditions). – [lulu](#) May 14, 2020 at 16:11

When you multiply M^{24} by your initial vector which has G_1 in the second position, the value of G_{24+1} will appear in the second position. Use that component instead of the first one, which as pointed out above is actually G_{24+2} . – [Oscar Lanzi](#) May 14, 2020 at 16:15

@lulu I now understand your point about the norm > 1 for one of the roots. But I'm still a little confused—perhaps it's the notation. the roots we get from the equation $x^3 = x^2 + x + 1$ —they're actually eigenvalues (which I often annotate with λ), right? Then how would I go about getting the λ in your notation (because I don't think that's an eigenvalue...)? – [singularity](#) May 15, 2020 at 2:36

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1 Answer

Sorted by: Highest score (default)

The provided equation is wrong, as you should have

$$\begin{bmatrix} G_{k+2} \\ G_{k+1} \\ G_k \end{bmatrix} = M^k \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

which matches when $k = 0$. The computation of G_k from this can be done in several ways. One way is to diagonalize M , which would reduce computing M^k down to computing λ^k for each eigenvalue λ . The largest eigenvalue, the [supergolden ratio](#) ψ , then determines the asymptotic behavior

$$\lim_{n \rightarrow \infty} \frac{\ln(G_n)}{n} = \lim_{n \rightarrow \infty} \ln \frac{G_{n+1}}{G_n} = \ln(\psi) \simeq 3822450858$$

of the supergolden sequence. The characteristic polynomial of the matrix

$$\det(M - \lambda I) = \det \begin{bmatrix} 1 - \lambda & 1 & 1 \\ 1 & -\lambda & 0 \\ 0 & 1 & -\lambda \end{bmatrix} = 1 + \lambda + \lambda^2 - \lambda^3$$

matches the characteristic equation of the recurrence. The eigendecomposition is given by

$$M = Q \Lambda Q^{-1}$$

where

$$Q \simeq \begin{bmatrix} 3.38298 & -0.191488 - 0.508852i & -0.191488 + 0.508852i \\ 1.83929 & -0.419643 + 0.606291i & -0.419643 - 0.606291i \\ 1 & 1 & 1 \end{bmatrix}$$

are the eigenvectors and

$$\Lambda \simeq \begin{bmatrix} 1.83929 & 0 & 0 \\ 0 & -0.419643 + 0.606291i & 0 \\ 0 & 0 & -0.419643 - 0.606291i \end{bmatrix}$$

are the eigenvalues. Only 6 significant figures are shown here, but in order to compute M^{23} accurately enough, more figures are probably needed.

Without involving non-integral values, the computation of G_n can be done quickly using exponentiation by squaring. For G_{25} , we get

$$\begin{aligned}
\begin{bmatrix} G_{25} \\ G_{24} \\ G_{23} \end{bmatrix} &= \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}^{23} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \left(\begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}^2 \right)^{11} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \\
&= \begin{bmatrix} 2 & 2 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}^{11} \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix} = \left(\begin{bmatrix} 2 & 2 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}^2 \right)^5 \begin{bmatrix} 2 & 2 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix} \\
&= \begin{bmatrix} 7 & 6 & 4 \\ 4 & 3 & 2 \\ 2 & 2 & 1 \end{bmatrix}^5 \begin{bmatrix} 9 \\ 5 \\ 3 \end{bmatrix} = \left(\begin{bmatrix} 7 & 6 & 4 \\ 4 & 3 & 2 \\ 2 & 2 & 1 \end{bmatrix}^2 \right)^2 \begin{bmatrix} 7 & 6 & 4 \\ 4 & 3 & 2 \\ 2 & 2 & 1 \end{bmatrix} \begin{bmatrix} 9 \\ 5 \\ 3 \end{bmatrix} \\
&= \begin{bmatrix} 81 & 68 & 44 \\ 44 & 37 & 24 \\ 24 & 20 & 13 \end{bmatrix}^2 \begin{bmatrix} 105 \\ 57 \\ 31 \end{bmatrix} = \begin{bmatrix} 81 & 68 & 44 \\ 44 & 37 & 24 \\ 24 & 20 & 13 \end{bmatrix} \begin{bmatrix} 13745 \\ 7473 \\ 4063 \end{bmatrix} \\
&= \begin{bmatrix} 1800281 \\ 978793 \\ 532159 \end{bmatrix}
\end{aligned}$$

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edited Jun 9, 2020 at 13:42

answered May 15, 2020 at 13:36



Simply Beautiful Art

74.4k 12 122 277



I think there's a mistake in the first part of your answer - in particular the limit of $\frac{\ln(G_n)}{n}$. – user787418 Jun 9, 2020 at 8:39



Thanks for the catch. – Simply Beautiful Art Jun 9, 2020 at 13:43